

Anup Barman's

CP-Arsenal



AnupBarman



AnupBarman



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1 Setup**1.1 Linux Build**

```
{
  "cmd" : ["ulimit -s 268435456; g++ -std=c++20
           $file_name -o $file_base_name && timeout 4s
           ./ $file_base_name < input.txt > output.txt"],
  "selector" : "source.c++",
  "shell" : true,
  "working_dir" : "$file_path"
}
```

1.2 Windows Build

```
{
  "cmd" : [ "g++.exe", "-std=c++20", "${file}", "-o",
            "${file_base_name}.exe", "&&",
            "${file_base_name}.exe<input.txt>output.txt" ],
  "selector" : "source.cpp",
  "shell" : true,
  "working_dir" : "$file_path"
}
```

2 Code**2.1 2D-IMOS**

```
diff[u][l]++;
diff[u][r+1]--;
diff[d+1][l]--;
diff[d+1][r+1]++;
// then use 2d prefix sum
```

2.2 2D-Prefix-Sum

```
pf[i][j] += pf[i-1][j] + pf[i][j-1] - pf[i-1][j-1];
// retrieve pfsum from (u, l) to (d, r)
pf[d][r] - pf[u-1][r] - pf[d][l-1] + pf[u-1][l-1];
```

2.3 Aho Corasic

```
//number of occurrence of word in a text
const ll N = 1e6+10, A = 26;
ll trie[N][A], pos[N], slink[N], dp[N], tot = 1;
vector<int> order;
void initTrie(){
  order.clear();
  while(tot--){
    memset(trie[tot], 0, sizeof(trie[tot]));
  }
  memset(pos, 0, sizeof(pos));
  memset(slink, 0, sizeof(slink));
  memset(dp, 0, sizeof(dp)); tot = 1;
}
void addStr(string &s, int ind){
  ll u = 0;
  for(auto it: s){
    ll n = it-'a';
    if(trie[u][n]==0) trie[u][n] = tot++;
    u = trie[u][n];
  } pos[ind] = u;
}
void build(){
  queue<ll> q; q.push(0);
  while(!q.empty()){
    ll p = q.front(); q.pop();
    order.push_back(p);
    for(ll c = 0; c<A; c++){
      ll u = trie[p][c];
      if(!u) continue;
      q.push(u);
      if(!p) continue;
      ll v = slink[p];
      while(v && !trie[v][c]) v = slink[v];
      slink[u] = trie[v][c];
    }
  }
}
void trav(string &s){
  ll u = 0;
  for(char c: s){
    c-'a';
    while(u && !trie[u][c]) u = slink[u];
    u = trie[u][c]; dp[u]++;
  }
  reverse(order.begin(), order.end());
  for(auto u: order){
    dp[slink[u]]+=dp[u];
  }
}
void solve(){
  ll n; cin>>n;
  string text; cin>>text;
  string s;
  for(ll i = 0; i<n; i++){
    cin>>s; addStr(s, i);
  }
  build(); trav(text);
  for(ll i = 0; i<n; i++){
    cout<<dp[pos[i]]<<endl;
  }
}
int32_t main(){
  ios_base::sync_with_stdio(0);
  cin.tie(0);
  ll tc = 1;
  cin>>tc;
  for(ll i = 1; i<=tc; i++){
    cout<<"Case "<<i<<": \n";
    initTrie();
    solve();
  }
}
```

2.4 Articulation Point

```
int n;
vector<vector<int>> lst; // adjacency list of graph
vector<bool> vis;
vector<int> tin, low;
int timer;
void dfs(int u, int p = -1) {
    vis[u] = true;
    tin[u] = low[u] = timer++;
    int children = 0;
    for (int v : lst[u]) {
        if (v == p) continue;
        if (vis[v]) {
            low[u] = min(low[u], tin[v]);
        } else {
            dfs(v, u);
            low[u] = min(low[u], low[v]);
            if (low[v] >= tin[u] && p != -1)
                IS_CUTPOINT(u);
            ++children;
        }
    }
    // if no vertex below v can reach u or higher
    // removing u disconnects that subtree
    if (p == -1 && children > 1)
        IS_CUTPOINT(u);
}
void find_cutpoints() {
    timer = 0;
    vis.assign(n, false);
    tin.assign(n, -1);
    low.assign(n, -1);
    for (int i = 0; i < n; ++i) {
        if (!vis[i])
            dfs(i);
    }
}
```

2.5 BFS

```
vector<vector<int>> adj; // adjacency list
// representation
int n; // number of nodes
int s; // source vertex
void bfs() {
    queue<int> q;
    vector<int> d(n, p(n));
    vector<bool> used(n);
    q.push(s);
    used[s] = true;
    p[s] = -1;
    while (!q.empty()) {
        int v = q.front();
        q.pop();
        for (int u : adj[v]) {
            if (!used[u]) {
                used[u] = true;
                q.push(u);
                d[u] = d[v] + 1;
                p[u] = v;
            }
        }
    }
}
// retrieving shortest path
if (!used[u]) {
    cout << "No path!";
} else {
    vector<int> path;
    for (int v = u; v != -1; v = p[v])
        path.push_back(v);
    reverse(path.begin(), path.end());
    cout << "Path: ";
}
```

```
for (int v : path)
    cout << v << " ";
}
```

2.6 Bellman Ford

```
#define ll long long
#define INF 1e18
void solve() {
    int n, m, v;
    cin >> n >> m >> v; // n = nodes, m = edges, v =
    // source (0-indexed)
    vector<array<ll, 3>> edges(m); // each edge: {a, b,
    // cost}
    for (int i = 0; i < m; i++) cin >> edges[i][0] >>
    // edges[i][1] >> edges[i][2];
    vector<ll> d(n, INF);
    vector<int> p(n, -1);
    d[v] = 0;
    int x = -1;
    for (int i = 0; i < n; i++) {
        x = -1;
        for (auto& e : edges) {
            int a = e[0], b = e[1];
            ll cost = e[2];
            if (d[a] < INF && d[b] > d[a] + cost) {
                d[b] = max(-INF, d[a] + cost);
                p[b] = a;
                x = b;
            }
        }
    }
    if (x == -1) {
        cout << "No negative cycle from vertex " << v <<
        '\n';
        return;
    }
    int y = x;
    for (int i = 0; i < n; i++) y = p[y];
    vector<int> path;
    for (int cur = y;; cur = p[cur]) {
        path.push_back(cur);
        if (cur == y && path.size() > 1) break;
    }
    reverse(path.begin(), path.end());
    cout << "Negative cycle: ";
    for (int u : path) cout << u << ' ';
    cout << '\n';
}
```

2.7 Bridge Finding Algorithm

```
const int MX = 1e5 + 10;
int n, m, timer = 0;
vector<int> adj[MX];
vector<int> tin(MX, -1), low(MX, -1);
vector<bool> vis(MX, false);
void is_bridge(int u, int v) {
    // do something with the edge
}
void dfs(int u, int p = -1) {
    vis[u] = true;
    tin[u] = low[u] = timer++;
    for (int v : adj[u]) {
        if (v == p) continue;
        if (vis[v]) {
            low[u] = min(low[u], tin[v]);
        } else {
            dfs(v, u);
            low[u] = min(low[u], low[v]);
            if (low[v] > tin[u]) {
                is_bridge(u, v);
            }
        }
    }
}
```

```
}
}
}
```

2.8 Centroid Decomposition

```
vector<int> lst[MX];
bool dead[MX];
int sz[MX];
void get_sz(int u, int p) {
    sz[u] = 1;
    for (int v : lst[u]) {
        if (v != p && !dead[v]) {
            get_sz(v, u);
            sz[u] += sz[v];
        }
    }
}
int find_cen(int u, int p, int tot) {
    for (int v : lst[u]) {
        if (v != p && !dead[v] && sz[v] > tot / 2) {
            return find_cen(v, u, tot);
        }
    }
    return u;
}
void calc(int cen) {
    // work on centroid here
}
void decomp(int u) {
    get_sz(u, -1);
    int tot = sz[u];
    int cen = find_cen(u, -1, tot);
    calc(cen);
    dead[cen] = true;
    for (int v : lst[cen]) {
        if (!dead[v]) {
            decomp(v);
        }
    }
}
```

2.9 Co-Ordinate Compression

```
vector<int> pos;
sort(pos.begin(), pos.end());
pos.erase(unique(pos.begin(), pos.end()), pos.end());
// then lower_bound on this pos array to find the
// compressed co-ordinate
```

2.10 CoinChange(MinCoins)

```
const int INF = 1e9;
int min_coins(int target, vector<int>& coins) {
    // DP Definition: dp[i] = minimum coins needed to
    // make value i
    vector<int> dp(target + 1, INF);
    // Base Case: 0 coins are needed to make value 0
    dp[0] = 0;
    for (int i = 1; i <= target; i++) {
        for (int c : coins) {
            if (i - c >= 0) {
                // Transition: Try taking coin 'c', adding 1 to
                // the count of (i-c)
                dp[i] = min(dp[i], dp[i - c] + 1);
            }
        }
    }
    return (dp[target] == INF) ? -1 : dp[target];
}
```

2.11 CoinChange(NumberOfWays)

```

ll count ways(int target, vector<int>& coins) {
    // DP Definition: dp[i] = number of ways to make
    // value i using given coins
    vector<ll> dp(target + 1, 0);
    // Base Case: There is 1 way to make value 0 (by
    // choosing nothing)
    dp[0] = 1;
    for (int c : coins) { // Loop coins *outside* to
        // count combinations (unordered)
        for (int i = c; i <= target; i++) {
            // Transition: Add ways to make the remainder (i
            // - c)
            dp[i] += dp[i - c];
        }
    }
    return dp[target];
}

```

2.12 CountSubsetWithSumK

```

int memo[105][10005]; // (N, Sum)
vector<int> nums;
int K;
int solve(int idx, int sum) {
    // DP: dp[idx][sum] = number of ways to complete the
    // sum using suffix nums[idx...]
    // Base: End of array. Return 1 if sum matches K,
    // else 0
    if (idx == nums.size()) {
        return sum == K;
    }
    if (memo[idx][sum] != -1) return memo[idx][sum];
    // Trans: Sum of ways (Include current element) +
    // (Exclude current element)
    int take = (sum + nums[idx] <= K) ? solve(idx + 1,
        sum + nums[idx]) : 0;
    int skip = solve(idx + 1, sum);
    return memo[idx][sum] = take + skip;
}

```

2.13 Cycle Detection in DAG

```

const int MX = 1e5 + 10;
bool vis[MX], pathVis[MX];
vector<int> lst[MX];
bool dfs(int u) {
    vis[u] = true;
    pathVis[u] = true;
    for (auto v : lst[u]) {
        if (!vis[v]) {
            if (dfs(v))
                return true;
        } else if (pathVis[v]) {
            return true;
        }
    }
    pathVis[u] = false;
    return false;
}
void solve() {
    // take graph input
    for (int i = 0; i < n; ++i) {
        if (!vis[i])
            dfs(i);
    }
}

```

2.14 DSU on Trees

```

int n, color[MX], ans[MX];
vector<int> g[MX];
set<int> bucket[MX];
int merge(int a, int b) {

```

```

    if (bucket[a].size() < bucket[b].size()) swap(a, b);
    bucket[a].insert(bucket[b].begin(), bucket[b].end());
    bucket[b].clear();
    return a;
}
int dfs(int u, int p = -1) {
    int cur = u;
    for (int v : g[u])
        if (v != p)
            cur = merge(cur, dfs(v, u));
    ans[u] = (int)bucket[cur].size();
    return cur;
}
void solve() {
    cin >> n;
    for (int i = 0; i < n; ++i) {
        cin >> color[i];
        bucket[i].insert(color[i]);
    }
    // graph input
    dfs(0);
    // print output
}

```

2.15 DSU

```

const int MX = 1e5 + 10;
int par[MX], sz[MX];
void init() {
    for (int i = 0; i < MX; i++) {
        par[i] = i;
        sz[i] = 1;
    }
}
int findpar(int x) {
    if (par[x] == x) return x;
    return par[x] = findpar(par[x]);
}
void unite(int u, int v) {
    u = findpar(u);
    v = findpar(v);
    if (u != v) {
        if (sz[u] < sz[v]) {
            swap(u, v);
        }
        sz[u] += sz[v];
        par[v] = u;
    }
}

```

2.16 Digit DP

```

string S;
ll memo[20][2][180];
ll dp(int idx, bool tight, int current_sum) {
    // DP Definition:
    // dp(idx, tight, current_sum) returns the count of
    // valid suffixes we can form
    // starting from index 'idx', given the 'tight'
    // constraint and the current state (sum).
    // Base Case: We have placed digits for all positions
    if (idx == S.size()) {
        return 1; // Found 1 valid number (or check:
        // return current_sum == target ? 1 : 0)
    }
    ll &ans = memo[idx][tight][current_sum];
    if (ans != -1) {
        return ans;
    }
    ll ans = 0;
    // Determine the upper bound for the current digit
    int limit = tight ? (S[idx] - '0') : 9;
    for (int digit = 0; digit <= limit; digit++) {
        // Transition:
        // Move to next index (idx + 1).

```

```

        // New tight constraint: (tight && (digit ==
        // limit))
        // New state: Update current_sum (or other state
        // variable)
        bool next_tight = tight && (digit == limit);
        // Example logic: filtering or constraints go here
        // if (digit == 4) continue; // Example: skip digit
        // 4
        ans += dp(idx + 1, next_tight, current_sum +
        // digit);
    }
    return memo[idx][tight][current_sum] = ans;
}
// Wrapper function to query for range [0, N]
ll query(ll N) {
    if (N < 0) return 0;
    if (N == 0) return 1; // Handling 0 case if needed
    S = to_string(N);
    memset(memo, -1, sizeof(memo));
    // Start from index 0, tight constraint is initially
    // TRUE, initial sum is 0
    return dp(0, true, 0);
}
// To get answer for range [L, R]: query(R) - query(L -
// 1)

```

2.17 Dijkstra

```

const int N = 1e5 + 5, INF = 1e18 + 7;
vector<pair<int, int>> g[N];
bool visited[N];
vector<int> dist(N, INF), parent(N);
bool dijkstra(int source) {
    priority_queue<pair<int, int>, vector<pair<int,
    // int>>, greater<pair<int, int>>> pq;
    pq.push({0, source});
    dist[source] = 0;
    parent[source] = -1;
    while (pq.size()) {
        int x = pq.top().second;
        pq.pop();
        if (visited[x]) continue;
        visited[x] = 1;
        for (auto [child_x, child_wt] : g[x]) {
            if (dist[x] + child_wt < dist[child_x]) {
                parent[child_x] = x;
                dist[child_x] = child_wt + dist[x];
                pq.push({dist[child_x], child_x});
            }
        }
    }
    return (dist[n] == INF);
}

```

2.18 Divisor Sieve

```

const int mxN = 1e5 + 10;
vector<int> divisors[mxN];
void divisorSieve() {
    for (int i = 1; i < mxN; i++) {
        for (int j = i; j < mxN; j += i) {
            divisors[j].push_back(i);
        }
    }
}

```

2.19 EditDistance

```
int edit_distance(string s1, string s2) {
    int m = s1.size(), n = s2.size();
    // DP: dp[i][j] = min ops to convert s1[0...i-1] to
    //       s2[0...j-1]
    vector<vector<int>> dp(m + 1, vector<int>(n + 1));
    // Base: Converting empty string to s2 requires j
    //       inserts (and vice versa)
    for (int i = 0; i <= m; i++) dp[i][0] = i;
    for (int j = 0; j <= n; j++) dp[0][j] = j;
    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1];
            } else {
                // Trans: 1 + min(Delete, Insert, Replace)
                dp[i][j] = 1 + min({dp[i - 1][j], dp[i][j - 1], dp[i - 1][j - 1]});
            }
        }
    }
    return dp[m][n];
}
```

2.20 Euler Tour

```
const int MX = 2e5 + 10;
int timer = -1;
// s = start pos, e = end pos
int val[MX], s[MX], e[MX], flat[MX];
vector<int> lst[MX];
void dfs(int u, int p) {
    s[u] = ++timer;
    flat[timer] = val[u];
    for (auto v : lst[u]) {
        if (v != p)
            dfs(v, u);
    }
    e[u] = timer;
}
```

2.21 Extended-Euclid

```
// Finds the greatest common divisor of a and b.
// Also finds integers x and y such that: ax + by =
// gcd(a, b)
int extendedGCD(int a, int b, int& x, int& y) {
    if (a == 0) {
        x = 0;
        y = 1;
        return b;
    }
    int x1, y1;
    int g = extendedGCD(b % a, a, x1, y1);
    x = y1 - (b / a) * x1;
    y = x1;
    return g;
}
```

2.22 Fast-Unordered-Map

```
mp.reserve(1 << 20); // about 1M buckets
mp.max_load_factor(0.7);
```

2.23 Floyd Warshall

```
vector<vector<int>> d(n, vector<int>(n, INF));
// take graph input into d
for (int k = 0; k < n; ++k) {
    for (int i = 0; i < n; ++i) {
        for (int j = 0; j < n; ++j) {
            if (d[i][k] < INF && d[k][j] < INF)
                d[i][j] = min(d[i][j], d[i][k] + d[k][j]);
        }
    }
}
```

```
}
}
}

2.24 Gosper-Hack
// Generates a single level r from nCr
vector<int> gen(int n, int r) {
    vector<int> v;
    int mask = (1 << r) - 1;
    do {
        v.push_back(mask);
        int x = mask & -mask;
        int y = mask + x;
        mask = ((mask & ~y) / x >> 1) | y;
    } while (mask < (1 << n));
    return v;
}
```

2.25 Inverse_Mod

```
// to calculate a-1 mod M
modpow(a, M - 2)
```

2.26 Is Sorted

```
is_sorted(first, last, comp);
```

2.27 KMP

```
vector<int> kmp(const string& s) {
    int n = (int)s.length();
    vector<int> pi(n);
    for (int i = 1; i < n; i++) {
        int j = pi[i - 1];
        while (j > 0 && s[i] != s[j])
            j = pi[j - 1];
        if (s[i] == s[j])
            j++;
        pi[i] = j;
    }
    return pi;
}
```

2.28 LCA using Binary Lifting

```
int n, l;
vector<vector<int>> adj;
int timer;
vector<int> tin, tout;
vector<vector<int>> up;
void dfs(int v, int p) {
    tin[v] = ++timer;
    up[v][0] = p;
    for (int i = 1; i <= l; ++i)
        up[v][i] = up[up[v][i - 1]][i - 1];
    for (int u : adj[v]) {
        if (u != p)
            dfs(u, v);
    }
    tout[v] = ++timer;
}
bool is_ancestor(int u, int v) {
    return tin[u] <= tin[v] && tout[u] >= tout[v];
}
int lca(int u, int v) {
    if (is_ancestor(u, v))
        return u;
    if (is_ancestor(v, u))
        return v;
    for (int i = l; i >= 0; --i) {
        if (!is_ancestor(up[u][i], v))
            u = up[u][i];
    }
```

```
}
return up[u][0];
}

void preprocess(int root) {
    tin.resize(n);
    tout.resize(n);
    timer = 0;
    l = ceil(log2(n));
    up.assign(n, vector<int>(l + 1));
    dfs(root, root);
}
```

2.29 LCS for 3 Strings

```
string a, b, c;
ll dp[55][55][55];
ll lcs(ll i, ll j, ll k) {
    if (i == a.size() or j == b.size() or k == c.size())
        return 0;
    if (dp[i][j][k] != -1) return dp[i][j][k];
    if (a[i] == b[j] and a[i] == c[k]) return 1 + lcs(i + 1, j + 1, k + 1);
    ll ans = 0;
    ans = max(ans, lcs(i, j, k + 1));
    ans = max(ans, lcs(i, j + 1, k));
    ans = max(ans, lcs(i + 1, j, k));
    return dp[i][j][k] = ans;
}
```

2.30 LIS

```
void LIS(const vector<int>& nums) {
    int n = nums.size();
    if (n == 0) return;
    // parent[i] stores the index of the predecessor of
    // nums[i] in the LIS
    vector<int> parent(n, -1);
    // tails id[k] stores the index (in nums) of the
    // smallest tail of all increasing subsequences of
    // length k+1
    vector<int> tails_id;
    // tails val[k] stores the actual value of that tail
    // (helper for binary search)
    vector<int> tails_val;
    // DP Definition (Implicit in greedy approach):
    // L[i] = Length of the longest increasing
    //       subsequence ending at index i.
    // We don't store L[i] explicitly, but compute it on
    // the fly using the 'tails' arrays.
    // Base Case:
    // The first element starts a subsequence of length
    // 1.
    for (int i = 0; i < n; i++) {
        int x = nums[i];
        // Find the first element in tails_val that is >= x
        auto it = lower_bound(tails_val.begin(), tails_val.end(), x);
        int idx = it - tails_val.begin();
        // Transition:
        // If x extends the longest existing subsequence,
        // append it.
        // Otherwise, replace the existing tail to maintain
        // the "smallest ending value" property
        // which allows future elements to extend the list
        // easier.
        // We also link nums[i] to the element currently
        // ending the subsequence of length (idx).
        if (it == tails_val.end()) {
            tails_val.push_back(x);
            tails_id.push_back(i);
        } else {
            tails_val[idx] = x;
            tails_id[idx] = i;
        }
    }
}
```



```

        *it = x;
        tails_id[idx] = i;
    }
    // If we are extending a sequence of length > 0,
    // the parent is the end of the previous length
    if (idx > 0) {
        parent[i] = tails_id[idx - 1];
    }
}
// Reconstruction
int curr = tails_id.back(); // Index of the last
// element of the LIS
vector<int> lis_path;
while (curr != -1) {
    lis_path.push_back(nums[curr]);
    curr = parent[curr];
}
reverse(lis_path.begin(), lis_path.end());
// Output
cout << "LIS Length: " << lis_path.size() << endl;
for (int x : lis_path) cout << x << " ";
cout << endl;
}

```

2.31 Lucas

```

int lucas(int n, int r) {
    int ans = 1;
    while (n > 0 or r > 0) {
        ans = ans * 1LL * nCr(n % M, r % M) % M;
        n /= M, r /= M;
    }
    return ans;
}

```

2.32 MST(Dense Graph)

```

int n;
const int MX = 2e3 + 5;
const int inf = 2e9;
vector<int> adj[MX];
int prim() {
    int total_weight = 0;
    vector<bool> selected(n, false);
    // min w[i] stores the minimum weight to reach node i
    vector<int> min_w(n, inf);
    // par[i] stores the node that connects to i
    // (previously min e[i].to)
    vector<int> par(n, -1);
    min_w[0] = 0;
    for (int i = 0; i < n; ++i) {
        int u = -1; // u is the Source node we are
        // selecting
        // Find the unselected node with the smallest min_w
        for (int j = 0; j < n; ++j) {
            if (!selected[j] and (u == -1 || min_w[j] <
                min_w[u])) {
                u = j;
            }
        }
        if (min_w[u] == inf) {
            cout << "No MST!" << "\n";
            exit(0);
        }
        selected[u] = true;
        total_weight += min_w[u];
        // Print edge u and the node that connected to it)
        if (par[u] != -1) {
            cout << u << " " << par[u] << "\n";
        }
        // Update neighbors
        // v is the Destination node (neighbor)
        for (int v = 0; v < n; ++v) {

```

```

            // Check if edge u -> v is shorter than v's
            // current best weight
            if (adj[u][v] < min_w[v]) {
                min_w[v] = adj[u][v];
                par[v] = u; // Store that we reached v from u
            }
        }
    }
    return total_weight;
}

```

2.33 MST(Sparse Graph)

```

// DSU first
void solve() {
    int n, m;
    cin >> n >> m;
    vector<array<int, 3>> edges;
    for (int i = 0; i < m; ++i) {
        int u, v, wt;
        cin >> u >> v >> wt;
        edges.push_back({wt, u, v});
    }
    sort(edges.begin(), edges.end());
    init(n);
    int cost = 0;
    for (auto &[wt, u, v] : edges) {
        if (findpar(u) == findpar(v)) continue;
        unite(u, v);
        cost += wt;
    }
    cout << cost << endl;
}

```

2.34 Manacher

```

// pal[1][i] = longest odd (half rounded down)
// palindrome around pos i and
// starts at i - pal[1][i] and ends at i + pal[1][i]
// pal[0][i] = half length of
// longest even palindrome around pos i, i + 1 and
// starts at i - par[0][i] + 1
// and ends at i + pal[0][i]
const int N = 5e5 + 10;
int pal[2][N];
void manacher(string& s) {
    int n = s.size(), idx = 2;
    while (idx-- > 0) {
        for (int l = -1, r = -1, i = 0; i < n - 1; ++i) {
            if (i > r)
                l = r = i;
            else {
                int k = min(r - i, pal[idx][l + r - i]);
                l = i - k, r = i + k;
            }
            while (l - idx >= 0 and r + 1 < n and s[l - idx]
                == s[r + 1]) l--, r++;
            pal[idx][i] = r - i;
            // [l - 1 + idx : r] palindrome
        }
    }
}

```

2.35 Matrix Exponentiation

```

const ll mod = 1e9;
vector<vector<ll>> matMul(vector<vector<ll>>& a,
    vector<vector<ll>>& b) {
    ll row1 = a.size(), col1 = a[0].size();
    ll row2 = b.size(), col2 = b[0].size();
    vector<vector<ll>> res(row1, vector<ll>(col2, 0));
    for (ll i = 0; i < row1; i++) {
        for (ll j = 0; j < col2; j++) {
            for (ll k = 0; k < row2; k++) {
                res[i][j] = (res[i][j] + (1LL * a[i][k] *
                    b[k][j]) % mod) % mod;
            }
        }
    }
}

```

```

    }
}
}
return res;
}
vector<vector<ll>> matExpo(vector<vector<ll>>& Mat, ll
    exp) {
    ll row = Mat.size(), col = Mat[0].size();
    ll p = row;
    vector<vector<ll>> res(p, vector<ll>(p, 0));
    for (ll i = 0; i < p; i++) res[i][i] = 1;
    while (exp) {
        if (exp & 1) res = matMul(res, Mat);
        Mat = matMul(Mat, Mat);
        exp >>= 1;
    }
    return res;
}
// b = (A(i), A(i-1), A(i-2), A(i-3))
// M = Magic matrix, nth = nth term, known = known
// value
ll get_nth(ll nth, ll known, vector<ll>& b,
    vector<vector<ll>>& M) {
    if (nth <= known) return b[nth - 1] % mod;
    reverse(b.begin(), b.end());
    vector<vector<ll>> me = matExpo(M, nth - known); //
    // MAT^(nth-known)
    ll ans = 0;
    for (int i = 0; i < known; i++) {
        ans = (ans + (b[i] * me[i][0]) % mod) % mod;
    }
    return ans;
}

```

2.36 Matrix Rotation

```

//90* clock-wise
now = {{0, 1, 0}, {-1, 0, 0}, {0, 0, 1}};
//90* anti-clock
now = {{0, -1, 0}, {1, 0, 0}, {0, 0, 1}};
//mirror with x axis at point p
now = {{-1, 0, 2 * p}, {0, 1, 0}, {0, 0, 1}};
//mirror with y axis at point p
now = {{1, 0, 0}, {0, -1, 2 * p}, {0, 0, 1}};
op[i + 1] = matMul(now, op[i]); // this
// op[i + 1] = matMul(op[i], now); //not this

```

2.37 Max Bipartite Matching [Hopcroft-Karp]

```

const int INF = 1e9;
void hopcroftCarp() {
    int n, m, e;
    cin >> n >> m >> e;
    vector<int> adj[n];
    for (int i = 0; i < e; ++i) {
        int u, v;
        cin >> u >> v;
        --u;
        adj[u].push_back(v);
    }
    vector<int> ml(m, -1), mr(n, -1), dist(n);
    auto bfs = [&]() -> bool {
        queue<int> q;
        for (int u = 0; u < n; ++u) {
            if (mr[u] == -1) {
                dist[u] = 0;
                q.push(u);
            } else {
                dist[u] = INF;
            }
        }
        bool foundAugmenting = false;
        while (!q.empty()) {
            int u = q.front();

```

```

q.pop();
for (int v : adj[u]) {
    int pairedLeft = ml[v];
    if (pairedLeft == -1) {
        foundAugmenting = true;
    } else if (dist[pairedLeft] == INF) {
        dist[pairedLeft] = dist[u] + 1;
        q.push(pairedLeft);
    }
}
return foundAugmenting;
};
function<bool(int)> dfs = [&](int u) -> bool {
    for (int v : adj[u]) {
        int pairedLeft = ml[v];
        if (pairedLeft == -1 or (dist[pairedLeft] ==
            dist[u] + 1 and dfs(pairedLeft))) {
            mr[u] = v;
            ml[v] = u;
            return true;
        }
    }
    dist[u] = INF;
    return false;
};
int matching = 0;
while (bfs()) {
    for (int u = 0; u < n; ++u) {
        if (mr[u] == -1) {
            if (dfs(u)) matching++;
        }
    }
}
cout << matching << endl;
for (int u = 0; u < n; ++u) {
    if (mr[u] != -1) {
        cout << u << " " << mr[u] << endl;
    }
}
}

```

2.38 Max Bipartite Matching [Kuhn's]

```

// left set size, right set size, edge count
int n, k, m, visToken = 1;
vector<int> lst[MX];
int mr[MX], ml[MX], vis[MX];
bool try_kuhn(int u) {
    if (vis[u] == visToken)
        return false;
    vis[u] = visToken;
    for (auto v : lst[u]) {
        if (ml[v] == -1 or try_kuhn(ml[v])) {
            ml[v] = u;
            mr[u] = v;
            return true;
        }
    }
    return false;
}
void solve() {
    cin >> n >> k >> m;
    for (int i = 0; i < m; ++i) {
        int u, v;
        cin >> u >> v;
        lst[u].push_back(v);
    }
    fill(mr, mr + n, -1);
    fill(ml, ml + k, -1);
    int ans = 0;
    for (int u = 0; u < n; ++u) {
        for (auto v : lst[u]) {
            if (ml[v] == -1) {
                ml[v] = u;
                mr[u] = v;
            }
        }
    }
}

```

```

ans++;
break;
}
}
for (int u = 0; u < n; ++u) {
    if (mr[u] != -1) continue;
    visToken++;
    if (try_kuhn(u))
        ans++;
}
cout << ans << endl;
for (int v = 0; v < k; ++v) {
    if (ml[v] != -1) {
        cout << ml[v] + 1 << " " << v + 1 << endl;
    }
}
}

```

2.39 Max Flow

```

struct Dinic {
    struct Edge {
        int to;
        ll capacity;
        int rev; // index of reverse edge
    };
    vector<vector<Edge>> adj;
    vector<int> level;
    vector<int> ptr;
    int n;
    Dinic(int nodes) : n(nodes), adj(nodes),
        level(nodes), ptr(nodes) {}
    void add_edge(int from, int to, ll cap) {
        adj[from].push_back({to, cap,
            (int)adj[to].size()});
        adj[to].push_back({from, 0, (int)adj[from].size()
            - 1});
    }
    bool bfs(int s, int t) {
        fill(level.begin(), level.end(), -1);
        level[s] = 0;
        queue<int> q;
        q.push(s);
        while (!q.empty()) {
            int v = q.front();
            q.pop();
            for (auto& edge : adj[v]) {
                if (edge.capacity > 0 && level[edge.to] == -1) {
                    level[edge.to] = level[v] + 1;
                    q.push(edge.to);
                }
            }
        }
        return level[t] != -1;
    }
    ll dfs(int v, int t, ll pushed) {
        if (pushed == 0) return 0;
        if (v == t) return pushed;
        for (int& cid = ptr[v]; cid < adj[v].size();
            ++cid) {
            auto& edge = adj[v][cid];
            int tr = edge.to;
            if (level[v] + 1 != level[tr] || edge.capacity
                == 0) continue;
            ll tr_pushed = dfs(tr, t, min(pushed,
                edge.capacity));
            if (tr_pushed == 0) continue;
            edge.capacity -= tr_pushed;
            adj[tr][edge.rev].capacity += tr_pushed;
            return tr_pushed;
        }
    }
}

```

```

}
return 0;
}
ll max_flow(int s, int t) {
    ll flow = 0;
    while (bfs(s, t)) {
        fill(ptr.begin(), ptr.end(), 0);
        while (ll pushed = dfs(s, t, 1e18)) {
            flow += pushed;
        }
    }
    return flow;
}
};
// Usage in int main():
int n, m;
cin >> n >> m;
Dinic dinic(n + 1); // Initialize with N+1 nodes (0 to
    N)
for (int i = 0; i < m; i++) {
    int u, v; ll cap;
    cin >> u >> v >> cap;
    // Add directed edge u -> v with capacity
    // For bidirectional, add edge v -> u as well
    dinic.add_edge(u, v, cap);
}
// Compute flow from node 1 (Source) to node n (Sink)
cout << dinic.max_flow(1, n) << endl;

```

2.40 Max Subarray Size Sum equal K

```

//write gpHashTable code before this part
void solution() {
    int n, k; cin >> n >> k;
    int total_sum = 0;
    vector<int> pre(n + 7, 0);
    for (int i = 1; i <= n; i++) {
        int temp; cin >> temp;
        total_sum += temp;
        if (i == 1) pre[i] = temp;
        else pre[i] = pre[i - 1] + temp;
    }
    if (total_sum < k) {
        cout << "-1" << endl; return;
    }
    if (total_sum == k) {
        cout << "0" << endl; return;
    }
    int maximum_subSize = 0;
    gp_hash_table<int, int, customHash> table;
    for (int i = 1; i <= n; i++) {
        if (pre[i] >= k) {
            int subSUM = pre[i] - k;
            if (subSUM == 0) {
                maximum_subSize = max(maximum_subSize, i);
            }
            else if (table[subSUM]) {
                int left = table[subSUM];
                int right = i; int subSize = right - left;
                maximum_subSize = max(subSize, maximum_subSize);
            }
        }
        if (!table[pre[i]]) table[pre[i]] = i;
    }
    cout << maximum_subSize << endl;
}

```

2.41 Maximum Subarray Sum(Kadanes)

```

ll max_subarray_sum(const vector<int>& nums) {
    if (nums.empty()) return 0;
    ll current_max = nums[0];
    ll global_max = nums[0];
    for (size_t i = 1; i < nums.size(); i++) {
        current_max = max((ll)nums[i], current_max +
            nums[i]);
    }
}

```

```

    global_max = max(global_max, current_max);
}
return global_max;
}

```

2.42 Merge Sort

```

// use array of elements, if multiple testcase make inv
// int inv = 0;
void merge(int vct[], int l, int m, int r) {
    int left = m - l + 1, right = r - m, lv[left],
        rv[right];
    for (int i = 0; i < left; i++) {
        lv[i] = vct[l + i];
    }
    for (int i = 0; i < right; i++) {
        rv[i] = vct[m + 1 + i];
    }
    int i = 0, j = 0, to = l;
    while (i < left && j < right) {
        if (lv[i] <= rv[j]) {
            vct[to] = lv[i];
            i++;
        } else {
            vct[to] = rv[j];
            j++;
        }
        // inversion count
        // int pore = left - i; inv += pore;
        to++;
    }
    while (i < left) {
        vct[to] = lv[i];
        i++;
        to++;
    }
    while (j < right) {
        vct[to] = rv[j];
        j++;
        to++;
    }
}
void merge_sort(int vct[], int l, int r) {
    if (r <= l) return;
    int m = l + ((r - l) / 2);
    merge_sort(vct, l, m);
    merge_sort(vct, m + 1, r);
    merge(vct, l, m, r);
}

```

2.43 Mex of All Subarray

```

const int N = 1e5 + 9, inf = 1e9;
struct ST {
    int t[4 * N];
    ST() {}
    void build(int n, int b, int e) {
        t[n] = 0;
        if (b == e) {
            return;
        }
        int mid = (b + e) >> 1, l = n << 1, r = l | 1;
        build(l, b, mid);
        build(r, mid + 1, e);
        t[n] = min(t[l], t[r]);
    }
    void upd(int n, int b, int e, int i, int x) {
        if (b > i || e < i) return;
        if (b == e && b == i) {
            t[n] = x;
            return;
        }
        int mid = (b + e) >> 1, l = n << 1, r = l | 1;
        upd(l, b, mid, i, x);
        upd(r, mid + 1, e, i, x);
    }
}

```

```

    t[n] = min(t[l], t[r]);
}
int get_min(int n, int b, int e, int i, int j) {
    if (b > j || e < i) return inf;
    if (b >= i && e <= j) return t[n];
    int mid = (b + e) >> 1, l = n << 1, r = l | 1;
    int L = get_min(l, b, mid, i, j);
    int R = get_min(r, mid + 1, e, i, j);
    return min(L, R);
}
int get_mex(int n, int b, int e, int i) { // mex of
    // [i... cur_id] if (b == e) return b;
    int mid = (b + e) >> 1, l = n << 1, r = l | 1;
    if (t[l] >= i) return get_mex(r, mid + 1, e, i);
    return get_mex(l, b, mid, i);
}
}
int a[N], f[N];
int32_t main() {
    ios_base::sync_with_stdio(0);
    cin.tie(0);
    int n;
    cin >> n;
    for (int i = 1; i <= n; i++) {
        cin >> a[i];
        --a[i];
    }
    t.build(1, 0, n);
    set<array<int, 3>> seg; // for cur_id = i, [x[0]...
        // i], [x[0] + 1...i], ...[x[1]...i] has mex, ,
        // = x[2]
    for (int i = 1; i <= n; i++) {
        int x = a[i];
        int r = min(i - 1, t.get_min(1, 0, n, 0, x - 1));
        int l = t.get_min(1, 0, n, 0, x) + 1;
        if (l <= r) {
            auto it = seg.lower_bound({l, -1, -1});
            while (it != seg.end() && (*it)[1] <= r) {
                auto x = *it;
                it = seg.erase(it);
            }
        }
        t.upd(1, 0, n, x, i);
        for (int j = r; j >= l; j--) {
            int m = t.get_mex(1, 0, n, j);
            int L = max(l, t.get_min(1, 0, n, 0, m) + 1);
            f[m] = 1;
            seg.insert({L, j, m});
            j = L - 1;
        }
        int m = !a[i];
        seg.insert({1, i, m});
        f[m] = 1;
    }
    int ans = 0;
    while (f[ans]) ++ans;
    cout << ans + 1 << '\n';
    return 0;
}

```

2.44 Mobius Function

```

//FULL MÖBIUS FUNCTION TEMPLATE
//Möbius Sieve (O(N log log N))
//Computes:
//mu[n] -> Möbius value
//primes[] -> list of primes
//isPrime[]
#include <bits/stdc++.h>
using namespace std;
const int MAXN = 1000000;
int mu[MAXN + 1];
bool isPrime[MAXN + 1];
vector<int> primes;

```

```

void mobius_sieve() {
    for (int i = 1; i <= MAXN; i++)
        mu[i] = 1, isPrime[i] = true;
    for (int i = 2; i <= MAXN; i++) {
        if (isPrime[i]) {
            primes.push_back(i);
            for (int j = i; j <= MAXN; j += i) {
                isPrime[j] = false;
                mu[j] *= -1;
            }
            long long sq = 1LL * i * i;
            if (sq <= MAXN) {
                for (int j = sq; j <= MAXN; j += sq)
                    mu[j] = 0;
            }
        }
    }
    mu[1] = 1;
}
//Count Numbers n Coprime with x
long long count_coprime(long long n, int x) {
    long long ans = 0;
    for (int d = 1; d * d <= x; d++) {
        if (x % d == 0) {
            ans += 1LL * mu[d] * (n / d);
            if (d != x / d)
                ans += 1LL * mu[x / d] * (n / (x / d));
        }
    }
    return ans;
}
//Count Pairs (i, j) with gcd(i, j) = 1 (1 <= i, j <= n)
long long count_coprime_pairs(int n) {
    long long ans = 0;
    for (int d = 1; d <= n; d++) {
        long long t = n / d;
        ans += 1LL * mu[d] * t * t;
    }
    return ans;
}
//Count Coprime Pairs in an Array (ICPC CLASSIC)
//Problem:
//Count (i, j) such that gcd(a[i], a[j]) = 1.
long long count_array_coprime_pairs(vector<int>& a) {
    int maxA = *max_element(a.begin(), a.end());
    vector<int> cnt(maxA + 1, 0);
    for (int x : a)
        cnt[x]++;
    vector<int> freq(maxA + 1, 0);
    for (int i = 1; i <= maxA; i++) {
        for (int j = i; j <= maxA; j += i)
            freq[i] += cnt[j];
    }
    long long ans = 0;
    for (int d = 1; d <= maxA; d++) {
        if (mu[d] != 0)
            ans += 1LL * mu[d] * freq[d] * (freq[d] - 1)
                / 2;
    }
    return ans;
}
//Mobius Inversion (Template)
vector<long long> mobius_inversion(vector<long long>&
    & f, int n) {
    vector<long long> g(n + 1, 0);
    for (int i = 1; i <= n; i++) {
        for (int j = i; j <= n; j += i) {
            g[j] += mu[i] * f[j / i];
        }
    }
    return g;
}

```



```

}
//When to Use This Template
//Use Möbius when problem mentions:
//gcd = 1
//coprime pairs
//divisor sums
//inclusion-exclusion
//multiplicative functions
//Complexity Summary
//Task
//Time
//Sieve
//O(N log log N)
//Coprime count
//O(x)
//Pair counting
//O(N)
//Array coprime
//O(A log A)

//Contest Tips (IMPORTANT)
//Always precompute once
//Combine with prefix sums
//Watch out for overflow
// values are only {-1, 0, +1}

```

2.45 Number of Pairs with GCD equal g

```

/*a[i] <= 1e6
for all 1<=g<=n, how many pairs exist such that g =
  - gcd(a[i], a[j]);
complexity : nlogn
*/
ll n; cin >> n;
ll a[n + 1]; memset(cnt, 0, sizeof cnt);
for (ll i = 1; i <= n; i++) {cin >> a[i]; cnt[a[i]]++;}
ll gcd[n + 1]; memset(gcd, 0, sizeof gcd);
for (ll i = n; i >= 1; i--) {
    ll pair = 0, invalid_pair = 0;
    for (ll j = i; j <= n; j += i) {
        pair += cnt[j];
        invalid_pair += gcd[j];
    }
    pair = (pair * (pair - 1)) / 2;
    gcd[i] = pair - invalid_pair;
}
// how many pairs exist whose gcd is i

```

2.46 Number of Subarray Sum is K

```

//write gpHashTable code before this part
void solution(){
int n, k; cin >> n >> k;
int total_sum = 0;
vector<int> pre(n + 7, 0);
for (int i = 1; i <= n; i++) {
    int temp; cin >> temp;
    total_sum += temp;
    if (i == 1) pre[i] = temp;
    else pre[i] = pre[i - 1] + temp; }
int cnt_subarray = 0;
gp_hash_table<int, int, customHash> table;
table[0] = 1;
for (int i = 1; i <= n; i++) {
    cnt_subarray += table[pre[i] - k];
    table[pre[i]]++;
} cout << cnt_subarray << endl; }

```

2.47 PBDS

```

#include "ext/pb_ds/assoc_container.hpp"
#include "ext/pb_ds/tree_policy.hpp"
using namespace gnu_pbds;
template<class T>
using oset = tree<T, null_type, less<T>, rb_tree_tag,

```

```

tree_order_statistics_node_update>;
Note: Use less_equal for multiset like behaviour
Usage:
st.find_by_order(k) :: returns iterator of the k-th
  - smallest element
st.order_of_key(x) :: returns index of x (number of
  - elements less than x)

```

2.48 Phi(1 to N)

```

const int MX = 1e7 + 10;
int phi[MX];
void sieve() {
    for (int i = 0; i < MX; i++) phi[i] = i;
    for (int i = 2; i < MX; i++) {
        if (phi[i] == i) {
            for (int j = i; j < MX; j += i) {
                phi[j] -= phi[j] / i;
            }
        }
    }
}

```

2.49 Phi

```

int phi(int n) {
    int rt = sqrt(n);
    int result = n;
    for (int i = 2; i <= rt; i++) {
        if (n % i == 0) {
            while (n % i == 0) n /= i;
            result -= result / i;
        }
    }
    if (n > 1) result -= result / n;
    return result;
}

```

2.50 Print Binary

```

void printBin(ll n) {
    int msb = 63 - __builtin_clzll(n);
    for (int i = msb; i >= 0; i--) {
        cout << ((n >> i) & 1);
    }
    cout << "\n";
}

```

2.51 Range-xor-Trick

```

// XOR from 0 to x
int fx(int x) {
    if (x % 4 == 0) return x;
    if (x % 4 == 1) return 1;
    if (x % 4 == 2) return x + 1;
    return 0; // x % 4 == 3
}
int rangeXOR(int l, int r) {
    return fx(r) ^ fx(l - 1);
}

```

2.52 SCC

```

vector<bool> visited; // keeps track of which vertices
  - are already visited
// runs depth first search starting at vertex v.
// each visited vertex is appended to the output vector
  - when dfs leaves it.
void dfs(int v, vector<vector<int>> &adj,
  - vector<int> &output) {
    visited[v] = true;
    for (auto u : adj[v])
        if (!visited[u])
            dfs(u, adj, output);
    output.push_back(v);
}

```

```

}
// input: adj -- adjacency list of G
// output: components -- the strongly connected
  - components in G
// output: adj_cond -- adjacency list of G^SCC (by root
  - vertices)
void strongly_connected_components(vector<vector<int>>
  - &adj,
  - vector<vector<int>>
    - &components,
    - vector<vector<int>>
      - &adj_cond) {

    int n = adj.size();
    components.clear(), adj_cond.clear();
    vector<int> order; // will be a sorted list of G's
      - vertices by exit time
    visited.assign(n, false);
    // first series of depth first searches
    for (int i = 0; i < n; i++)
        if (!visited[i])
            dfs(i, adj, order);

    // create adjacency list of G^T
    vector<vector<int>> adj_rev(n);
    for (int v = 0; v < n; v++)
        for (int u : adj[v])
            adj_rev[u].push_back(v);
    visited.assign(n, false);
    reverse(order.begin(), order.end());
    vector<int> roots(n, 0); // gives the root vertex
      - of a vertex's SCC
    // second series of depth first searches
    for (auto v : order)
        if (!visited[v]) {
            std::vector<int> component;
            dfs(v, adj_rev, component);
            components.push_back(component);
            int root = *component.begin();
            for (auto u : component)
                roots[u] = root;
        }

    // add edges to condensation graph
    adj_cond.assign(n, {});
    for (int v = 0; v < n; v++)
        for (auto u : adj[v])
            if (roots[v] != roots[u])
                adj_cond[roots[v]].push_back(roots[u]);
}

```

2.53 SOD NOD

```

// SOD = ((P^(x+1)-1)/(P-1)) * ((Q^(y+1)-1)/(Q-1)) *
  - ((R^(z+1)-1)/(R-1))
// NOD = P^x * Q^y * R^z => here, P, Q, R are prime
  - factors & x, y, z are
// powers NOD = (x + 1) (y + 1) (z + 1)
pair<int, int> SOD_NOD(int n) {
    int sod = 1, nod = 1;
    for (int i = 2; i * i <= n; ++i) {
        if (n % i == 0) {
            int pown = 1, pows = 0;
            while (n % i == 0) {
                pown *= i; // p^e
                pows++;
                n /= i;
            }
            pown *= i;
            sod *= (pown - 1) / (i - 1); // p^e+1
            nod *= (pows + 1); // (p^e+1)-1 / p-1
        }
    }
}

```

```

    }
    if (n > 1) {
        sod *= (n + 1);
        nod *= 2;
    }
    return {sod, nod};
}

```

2.54 Segment Tree(BSUA)

```

// CSES - 1749
const int MX = 2e5 + 10;
int n;
int arr[MX], st[MX << 2];
void assign(int i, int x, int u = 1, int s = 0, int e
    = n - 1) {
    if (s == e) {
        st[u] = x;
        return;
    }
    int v = u << 1, w = v | 1, m = (s + e) >> 1;
    if (i <= m) assign(i, x, v, s, m);
    else assign(i, x, w, m + 1, e);
    st[u] = st[v] + st[w];
}
int kth(int k, int u = 1, int s = 0, int e = n - 1) {
    if (st[u] < k) return -1;
    if (s == e) {
        return s;
    }
    int v = u << 1, w = v | 1, m = (s + e) >> 1;
    if (st[v] >= k) return kth(k, v, s, m);
    else return kth(k - st[v], w, m + 1, e);
}
void solve() {
    cin >> n;
    for (int i = 0; i < n; ++i) {
        cin >> arr[i];
    }
    for (int i = 0; i < n; ++i) {
        assign(i, 1);
    }
    for (int i = 0; i < n; ++i) {
        int x;
        cin >> x;
        int ind = kth(x);
        assign(ind, 0);
        cout << arr[ind] << " ";
    }
}

```

2.55 Segment Tree(LzP)

```

class stree {
    vector<ll> st, lazy;
public:
    stree(int n) {
        st.assign((n << 2) + 10, 0);
        lazy.assign((n << 2) + 10, 0);
    }
    void push(int u, int s, int e) {
        if (!lazy[u]) return;
        st[u] += (e - s + 1) * 1LL * lazy[u];
        if (s != e) {
            int v = u << 1, w = v | 1, m = (s + e) >> 1;
            lazy[v] += lazy[u];
            lazy[w] += lazy[u];
        }
        lazy[u] = 0;
    }
    void build(int u, int s, int e, int arr[]) {
        if (s == e) {
            st[u] = arr[s];
            return;
        }
    }
}

```

```

int v = u << 1, w = v | 1, m = (s + e) >> 1;
build(v, s, m, arr);
build(w, m + 1, e, arr);
st[u] = st[v] + st[w];
}
void update(int l, int r, int x, int u, int s, int
    e) {
    push(u, s, e);
    if (e < l or r < s) return;
    int v = u << 1, w = v | 1, m = (s + e) >> 1;
    if (l <= s and e <= r) {
        st[u] += (e - s + 1) * 1LL * x;
        if (s != e) {
            lazy[v] += x;
            lazy[w] += x;
        }
        return;
    }
    update(l, r, x, v, s, m);
    update(l, r, x, w, m + 1, e);
    st[u] = st[v] + st[w];
}
ll query(int l, int r, int u, int s, int e) {
    push(u, s, e);
    if (e < l or r < s) return 0;
    if (l <= s and e <= r) return st[u];
    int v = u << 1, w = v | 1, m = (s + e) >> 1;
    return query(l, r, v, s, m) + query(l, r, w, m +
        1, e);
}
};

```

2.56 Segment Tree

```

struct stree {
    vector<ll> st;
    stree(int n) {
        st.assign((n << 2) + 10, 0);
    }
    void build(int u, int s, int e, int arr[]) {
        if (s == e) {
            st[u] = arr[s];
            return;
        }
        int v = u << 1, w = v | 1, m = (s + e) >> 1;
        build(v, s, m, arr);
        build(w, m + 1, e, arr);
        st[u] = st[v] + st[w];
    }
    ll query(int l, int r, int u, int s, int e) {
        if (e < l or r < s) return 0;
        if (l <= s and e <= r) return st[u];
        int v = u << 1, w = v | 1, m = (s + e) >> 1;
        return query(l, r, v, s, m) + query(l, r, w, m +
            1, e);
    }
    void update(int i, int x, int u, int s, int e) {
        if (s == e) {
            st[u] = x;
            return;
        }
        int v = u << 1, w = v | 1, m = (s + e) >> 1;
        if (i <= m)
            update(i, x, v, s, m);
        else
            update(i, x, w, m + 1, e);
        st[u] = st[v] + st[w];
    }
};

```

2.57 Segmented Sieve

```

void segSeive(ll low, ll high) {
    vector<bool> area((high - low) + 1, true);
    for (ll i = 0; primes[i] * primes[i] <= high; i++) {
        ll start = ((low / primes[i]) * primes[i]);
    }
}

```

```

if (start < low) start += primes[i];
for (ll j = start; j <= high; j += primes[i]) {
    if (j == primes[i]) continue;
    area[j - low] = false;
}
}
for (ll i = 0; i < (high - low) + 1; i++) {
    if (area[i]) {
        if (i + low != 1 and i + low != 0) {
            cout << i + low << endl;
        }
    }
}
}
}

```

2.58 Sparse-Table

```

const int MX = 2e5 + 10;
const int LOG = 25;
int arr[MX], st[MX][LOG];
void build(int n) {
    for (int i = 0; i < n; ++i) {
        st[i][0] = arr[i];
    }
    for (int j = 1; j < LOG; ++j) {
        for (int i = 0; i + (1 << j) <= n; ++i) {
            st[i][j] = min(st[i][j - 1], st[i + (1 << (j -
                1))] [j - 1]);
        }
    }
}
int query(int l, int r) {
    int len = r - l + 1;
    int k = lg(len);
    return min(st[l][k], st[r - (1 << k) + 1][k]);
}

```

2.59 String Hashing 2

```

const int N = 1000010, MOD = 1e9 + 7;
const ll P[] = {97, 1000003};
ll bigMod(ll a, ll e) {
    if (e == -1) e = MOD - 2;
    ll ret = 1;
    while (e) {
        if (e & 1) ret = ret * a % MOD;
        a = a * a % MOD; e >>= 1;
    }
    return ret;
}
ll pwr[2][N], inv[2][N];
void initHash() {
    for (int it = 0; it < 2; ++it) {
        pwr[it][0] = inv[it][0] = 1;
        ll INV_P = bigMod(P[it], -1);
        for (int i = 1; i < N; ++i) {
            pwr[it][i] = pwr[it][i - 1] * P[it] % MOD;
            inv[it][i] = inv[it][i - 1] * INV_P % MOD;
        }
    }
}
struct RangeHash {
    vector<ll> h[2], rev[2];
    RangeHash(const string S, bool revFlag = 0) {
        for (int it = 0; it < 2; ++it) {
            h[it].resize(S.size() + 1, 0);
            for (int i = 0; i < S.size(); ++i) {
                h[it][i + 1] = (h[it][i] + pwr[it][i + 1] *
                    (S[i] - 'a' + 1)) % MOD;
            }
            if (revFlag) {
                rev[it].resize(S.size() + 1, 0);
                for (int i = 0; i < S.size(); ++i) {
                    rev[it][i + 1] =
                }
            }
        }
    }
}

```

```

        (rev[it][i] + inv[it][i + 1] * (S[i] -
        'a' + 1)) % MOD;
    }
}
inline ll get(int l, int r) {
    ll one = (h[0][r + 1] - h[0][l]) * inv[0][l + 1] %
    MOD;
    ll two = (h[1][r + 1] - h[1][l]) * inv[1][l + 1] %
    MOD;
    if (one < 0) one += MOD;
    if (two < 0) two += MOD;
    return one << 31 | two;
}
inline ll getReverse(int l, int r) {
    ll one = (rev[0][r + 1] - rev[0][l]) * pwr[0][r +
    1] % MOD;
    ll two = (rev[1][r + 1] - rev[1][l]) * pwr[1][r +
    1] % MOD;
    if (one < 0) one += MOD;
    if (two < 0) two += MOD;
    return one << 31 | two;
}
};

```

2.60 String Hashing

```

const int mod1 = 911382323, mod2 = 972663749, b1 =
    137, b2 = 139;
const int mxN = 5000010;
int pow_b1[mxN], pow_b2[mxN], inv_b1[mxN], inv_b2[mxN];
int binExp(int base, int power, int mod) {
    int res = 1;
    while (power) {
        if (power & 1) res = (1LL * res * base) % mod;
        base = (1LL * base * base) % mod;
        power >>= 1;
    }
    return res;
}
void pre() {
    pow_b1[0] = pow_b2[0] = 1;
    for (int i = 1; i < mxN; i++) {
        pow_b1[i] = (1LL * pow_b1[i - 1] * b1) % mod1;
        pow_b2[i] = (1LL * pow_b2[i - 1] * b2) % mod2;
    }
    inv_b1[mxN - 1] = binExp(pow_b1[mxN - 1], mod1 - 2,
    mod1);
    inv_b2[mxN - 1] = binExp(pow_b2[mxN - 1], mod2 - 2,
    mod2);
    for (int i = mxN - 2; i >= 0; i--) {
        inv_b1[i] = (1LL * inv_b1[i + 1] * b1) % mod1;
        inv_b2[i] = (1LL * inv_b2[i + 1] * b2) % mod2;
    }
}
vector<pair<int, int>> getPref(string& s) {
    int qq = s.size();
    vector<pair<int, int>> hsh(qq);
    for (int i = 0; i < qq; i++) {
        if (i == 0) {
            hsh[i].first = (1LL * s[i] * pow_b1[i]) % mod1;
            hsh[i].second = (1LL * s[i] * pow_b2[i]) % mod2;
        } else {
            hsh[i].first =
                (hsh[i - 1].first + (1LL * s[i] * pow_b1[i])
                % mod1) % mod1;
            hsh[i].second =
                (hsh[i - 1].second + (1LL * s[i] *
                pow_b2[i]) % mod2) % mod2;
        }
    }
    return hsh;
}
pair<int, int> getHash(string& str) {

```

```

    int hsh1 = 0, hsh2 = 0, sz = str.size();
    for (int i = 0; i < sz; ++i) {
        hsh1 = (hsh1 + 1LL * str[i] * pow_b1[i] % mod1) %
        mod1;
    }
    for (int i = 0; i < sz; ++i) {
        hsh2 = (hsh2 + 1LL * str[i] * pow_b2[i] % mod2) %
        mod2;
    }
    return {hsh1, hsh2};
}
pair<int, int> getSub(int l, int r, vector<pair<int,
    int>>& v) {
    pair<int, int> q;
    if (l == 0) {
        q = {v[r].first, v[r].second};
    } else {
        int x = (1LL * ((v[r].first - v[l - 1].first +
        mod1) % mod1) * inv_b1[l]) %
        mod1;
        int y =
            (1LL * ((v[r].second - v[l - 1].second + mod2)
            % mod2) * inv_b2[l]) %
            mod2;
        q = {x, y};
    }
    return q;
}

```

2.61 Tonelli-Shanks

```

// Calculate X when we know Y and P for X^2 = Y (mod P)
ll tonelli_shanks(ll Y, ll P) {
    if (Y == 0) return 0;
    if (P == 2) return Y % 2;
    // Euler's criterion: Check if Y is a quadratic
    // residue
    if (modPow(Y, (P - 1) / 2, P) != 1) {
        return -1;
    }
    ll Q = P - 1;
    ll S = 0;
    while (Q % 2 == 0) {
        Q /= 2;
        S++;
    }
    ll z = 2;
    while (modPow(z, (P - 1) / 2, P) != P - 1) {
        z++;
    }
    ll M = S;
    ll c = modPow(z, Q, P);
    ll t = modPow(Y, Q, P);
    ll R = modPow(Y, (Q + 1) / 2, P);
    while (t != 0 && t != 1) {
        ll t2i = t;
        ll i = 0;
        for (i = 1; i < M; i++) {
            t2i = (t2i * t2i) % P;
            if (t2i == 1) break;
        }
        ll b = c;
        for (int j = 0; j < M - i - 1; j++) {
            b = (b * b) % P;
        }
        ll b2 = (b * b) % P;
        M = i;
        c = b2;
        t = (t * b2) % P;
        R = (R * b) % P;
    }
    return R;
}

```

2.62 Topological Sorting

```

const int MX = 1e5 + 10;
vector<int> lst[MX];
vector<int> topSort(int n) {
    int in[n];
    for (int u = 0; u < n; ++u) {
        for (int v : lst[u]) {
            in[v]++;
        }
    }
    queue<int> q;
    for (int u = 0; u < n; ++u) {
        if (in[u] == 0) {
            q.push(u);
        }
    }
    vector<int> order;
    while (!q.empty()) {
        int u = q.front();
        q.pop();
        order.push_back(u);
        for (int v : lst[u]) {
            in[v]--;
            if (in[v] == 0) {
                q.push(v);
            }
        }
    }
    // If the topological sort doesn't contain all nodes,
    // there is a cycle.
    if (order.size() < n) {
        return {}; // Return empty to indicate cycle
    }
    return order;
}

```

2.63 Unique Elements in vector

```

sort(v.begin(), v.end());
v.erase(unique(v.begin(), v.end()), v.end());

```

2.64 Unique Prime Factorization using Sieve

```

const int MX = 2e5 + 10;
vector<int> pfac[MX];
void factorize() {
    for (int i = 2; i < MX; i++) {
        if (!pfac[i].empty()) continue;
        for (int j = i; j < MX; j += i)
            pfac[j].push_back(i);
    }
}

```

2.65 Weighted Union Find

```

const int MX = 2e5 + 10;
int par[MX], sz[MX];
ll d[MX];
void init() {
    for (int i = 0; i < MX; ++i) {
        par[i] = i;
        sz[i] = 1;
        d[i] = 0;
    }
}
int findpar(int x) {
    if (par[x] == x) return x;
    int p = par[x];
    par[x] = findpar(p);
    d[x] += d[p];
    return par[x];
}
bool unite(int a, int b, ll w) {

```

```

int ra = findpar(a);
int rb = findpar(b);
if (ra == rb) {
    return (d[b] - d[a] == w);
}
if (sz[ra] < sz[rb]) {
    swap(a, b);
    swap(ra, rb);
    w = -w;
}
par[rb] = ra;
d[rb] = d[a] + w - d[b];
sz[ra] += sz[rb];
return true;
}
dist(int a, int b) {
    findpar(a), findpar(b);
    return d[b] - d[a];
}

```

2.66 erase-while-iteration

```

for (auto it = st.begin(); it != st.end(); ) {
    if (temp.find(*it) == temp.end()) {
        st.erase(it++);
    } else {
        ++it;
    }
}

```

2.67 int128

```

__int128 read() {
    __int128 x = 0, f = 1;
    char ch = getchar();
    while (ch < '0' || ch > '9') {
        if (ch == '-') f = -1;
        ch = getchar();
    }
    while (ch >= '0' && ch <= '9') {
        x = x * 10 + ch - '0';
        ch = getchar();
    }
    return x * f;
}
void print(__int128 x) {
    if (x < 0) {
        putchar('-');
        x = -x;
    }
    if (x > 9) print(x / 10);
    putchar(x % 10 + '0');
}

```

2.68 legendre

```

// returns power of prime p in n!
long long legendre(long long n, long long p) {
    long long ans = 0;
    while (n > 0) {
        n /= p;
        ans += n;
    }
    return ans;
}

```

2.69 nCr-and-nPr

```

const int MX = 1e6 + 10;
const int M = 1e9 + 7;
int fact[MX], inv_fact[MX];
// modpow here
void precalFact() {
    fact[0] = inv_fact[0] = 1;
    for (int i = 1; i < MX; i++) {
        fact[i] = (1LL * fact[i - 1] * i) % M;
    }
}

```

```

inv_fact[MX - 1] = modpow(fact[MX - 1], M - 2);
for (int i = MX - 2; i >= 1; i--) {
    inv_fact[i] = (1LL * inv_fact[i + 1] * (i + 1)) %
        M;
}
}
int nCr(int n, int r) {
    if (r < 0 or r > n) return 0;
    return 1LL * fact[n] * inv_fact[r] % M * inv_fact[n
        - r] % M;
}
int nPr(int n, int r) {
    if (n < r) return 0;
    return 1LL * fact[n] * inv_fact[n - r] % M;
}

```

3 Geometry

3.1 Convex Hull

```

vector<pt> convexHull(vector<pt> p) {
    sort(p.begin(), p.end());
    vector<pt> hull;
    for (int rep = 0; rep < 2; ++rep) {
        const int S = hull.size();
        for (pt c : p) {
            while ((int)hull.size() - S >= 2) {
                pt a = hull.end()[-2];
                pt b = hull.end()[-1];
                if (a.cross(b, c) <= 0) {
                    // b is on the left of c, it is valid
                    break;
                }
                hull.pop_back(); // remove b
            }
            hull.push_back(c);
        }
        hull.pop_back();
        reverse(p.begin(), p.end());
    }
    return hull;
}

```

3.2 Line Line Intersection

```

bool lineLineIntersection(P p1, P p2, P p3, P p4) {
    if ((p2 - p1).cross((p4 - p3)) == 0) {
        // lines are parallel
        // basically the slopes of two lines are same,
        // producing the cross product 0
        if (p1.cross(p2, p3) != 0) {
            // lines are not collinear
            return false;
        }
        // bounding box idea, 2 lines are collinear but not
        // intersecting
        for (int rep = 0; rep < 2; ++rep) {
            if (max(p1.x, p2.x) < min(p3.x, p4.x) or
                max(p1.y, p2.y) < min(p3.y, p4.y)) {
                return false;
            }
            swap(p1, p3), swap(p2, p4);
        }
        // lines are collinear and intersecting
        return true;
    }
    // lines are not parallel
    for (int rep = 0; rep < 2; ++rep) {
        ll cp1 = p1.cross(p2, p3);
        ll cp2 = p1.cross(p2, p4);
        if ((cp1 < 0 and cp2 < 0) or (cp1 > 0 and cp2 >
            0)) {
            // lines are not intersecting
            return false;
        }
        swap(p1, p3), swap(p2, p4);
    }
}

```

```

}
// lines are intersecting
return true;
}

```

3.3 Point in Polygon

```

bool segmentContains(pt a, pt b, pt c) {
    if (a.cross(b, c) != 0) {
        return false;
    }
    return (min(a.x, b.x) <= c.x and c.x <= max(a.x,
        b.x)) and (min(a.y, b.y) <= c.y and c.y <=
        max(a.y, b.y));
}
void solve() {
    int n;
    cin >> n;
    pt p[n];
    for (int i = 0; i < n; ++i) {
        cin >> p[i];
    }
    int q;
    cin >> q;
    while (q--) {
        pt temp;
        cin >> temp;
        int cnt = 0;
        bool flag = false;
        for (int i = 0; i < n; ++i) {
            int j = (i + 1) % n;
            if (segmentContains(p[i], p[j], temp)) {
                flag = true;
                break;
            }
            pt l, r;
            if (p[i].x <= p[j].x) {
                l = p[i];
                r = p[j];
            } else {
                l = p[j];
                r = p[i];
            }
            if (temp.cross(l, r) < 0) {
                if (l.x <= temp.x and temp.x < r.x) {
                    cnt++;
                }
            }
        }
        if (flag) {
            cout << "BOUNDARY\n";
        } else if (cnt % 2 == 0) {
            cout << "OUTSIDE\n";
        } else {
            cout << "INSIDE\n";
        }
    }
}

```

3.4 Polygon Area

```

// the given points needs to be adjacent
void polygonArea() {
    int n;
    cin >> n;
    P p[n];
    for (int i = 0; i < n; ++i) {
        cin >> p[i];
    }
    ll area = 0;
    for (int i = 0; i < n; ++i) {
        // calculating wrt the origin
    }
}

```



```
    area += p[i].cross(p[(i + 1) % n]);
}
cout << abs(area) << "\n";
}
```

3.5 Polygon Lattice Points

```
// using picks theorem
// interior = area - (boundary / 2) + 1
// 2 * interior = 2 * area - boundary + 2
void solve() {
    int n;
    cin >> n;
    pt p[n];
    for (int i = 0; i < n; ++i) {
        cin >> p[i];
    }
    ll boundary = 0, area = 0;
    for (int i = 0; i < n; ++i) {
        int j = (i + 1) % n;
        pt diff = p[j] - p[i];
        boundary += gcd(diff.x, diff.y);
        area += p[i].cross(p[j]);
    }
    area = abs(area);
    ll interior = (area - boundary + 2) / 2;
    cout << interior << " " << boundary << "\n";
}
```

3.6 Template

```
struct pt {
    ll x, y;
    friend istream& operator>>(istream& is, pt& p) {
        return is >> p.x >> p.y;
    }
    friend ostream& operator<<(ostream& os, const pt& p)
    {
        return os << p.x << " " << p.y;
    }
    pt operator-(const pt& other) const {
        return {x - other.x, y - other.y};
    }
    void operator+=(const pt& other) {
        x += other.x, y += other.y;
    }
    bool operator<(const pt& b) {
        return make_pair(x, y) < make_pair(b.x, b.y);
    }
    ll cross(const pt& other) const {
        return x * other.y - y * other.x;
    }
    ll cross(const pt& a, const pt& b) const {
        return (a - *this).cross(b - *this);
    }
};
```

4 Notes

4.1 Geometry

4.1.1 Triangles

Circumradius: $R = \frac{abc}{4A}$, Inradius: $r = \frac{A}{s}$
The area of a triangle using two sides and the included angle can be given as:

$$A = \frac{1}{2}ab \sin \angle C$$

Length of median (divides triangle into two equal-area triangles):
 $m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$

Length of bisector (divides angles in two): $s_a = \sqrt{bc \left[1 - \left(\frac{a}{b+c} \right)^2 \right]}$

Law of tangents: $\frac{a+b}{a-b} = \frac{\tan \frac{\alpha+\beta}{2}}{\tan \frac{\alpha-\beta}{2}}$

Area given 3 sides of a triangle a, b, c : $A = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{a+b+c}{2}$

4.1.2 Sphere

Volume of a sphere: $V = \frac{4}{3}\pi R^3$
Spherical Cap volume: $V(h) = \frac{\pi h^2(3R-h)}{3}$

4.1.3 Quadrilaterals

With side lengths a, b, c, d , diagonals e, f , diagonals angle θ , area A and magic flux $F = b^2 + d^2 - a^2 - c^2$:

$$4A = 2ef \cdot \sin \theta = F \tan \theta = \sqrt{4e^2f^2 - F^2}$$

For cyclic quadrilaterals the sum of opposite angles is 180° , $ef = ac + bd$, and $A = \sqrt{(p-a)(p-b)(p-c)(p-d)}$.

4.1.4 Spherical coordinates

$$\begin{aligned} x &= r \sin \theta \cos \phi & r &= \sqrt{x^2 + y^2 + z^2} \\ y &= r \sin \theta \sin \phi & \theta &= \arccos(z / \sqrt{x^2 + y^2 + z^2}) \\ z &= r \cos \theta & \phi &= \operatorname{atan2}(y, x) \end{aligned}$$

4.1.5 Pick's Theorem

Given a lattice polygon with non-zero area, we define: S as the area of the polygon, I as the number of integer-coordinate points strictly inside the polygon, B as the number of integer-coordinate points on the boundary of the polygon. Then, Pick's Theorem states:

$$S = I + \frac{B}{2} - 1$$

The number of lattice points on segments (x_1, y_1) to (x_2, y_2) is: $\gcd(|x_2 - x_1|, |y_2 - y_1|) + 1$

4.1.6 Polygon

For a regular polygon with n sides and side length a , the circumradius R is given by:

$$R = \frac{a}{2 \sin(\frac{\pi}{n})}$$

4.1.7 Area of a Circular Segment

The area of a circular segment, which is the region enclosed by a chord and the corresponding arc, can be calculated using the formula:

$$A = \frac{R^2}{2} (\theta - \sin \theta)$$

where: R is the radius of the circle, θ is the central angle subtended by the chord, in radians.

4.1.8 Matrix Rotation

Anti-Clockwise Rotation:

$$\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix}$$

Clockwise Rotation:

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

4.2 Number Theory

4.2.1 Primes

$p = 962592769$ is such that $2^{21} \mid p - 1$, which may be useful. For hashing use 970592641 (31-bit number), 31443539979727 (45-bit), 3006703054056749 (52-bit). There are 78498 primes less than 1 000 000.

Primitive roots exist modulo any prime power p^a , except for $p = 2, a > 2$, and there are $\phi(\phi(p^a))$ many. For $p = 2, a > 2$, the group $\mathbb{Z}_{2^a}^\times$ is instead isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_{2^{a-2}}$.

4.2.2 Estimates

$\sum_{d|n} d = O(n \log \log n)$.

The number of divisors of n is at most around 100 for $n < 5e4$, 500 for $n < 1e7$, 2000 for $n < 1e10$, 200 000 for $n < 1e19$.

4.2.3 Perfect numbers

$n > 1$ is called perfect if it equals sum of its proper divisors and 1. Even n is perfect iff $n = 2^{p-1}(2^p - 1)$ and $2^p - 1$ is prime (Mersenne's). No odd perfect numbers are yet found.

4.2.4 Carmichael numbers

A positive composite n is a Carmichael number ($a^{n-1} \equiv 1 \pmod n$) for all a : $\gcd(a, n) = 1$, iff n is square-free, and for all prime divisors p of n , $p - 1$ divides $n - 1$.

4.2.5 Totient

- If p is a prime $\phi(p^k) = p^k - p^{k-1}$
- If a & b are relatively prime, $\phi(ab) = \phi(a)\phi(b)$
- $\phi(n) = n(1 - \frac{1}{p_1})(1 - \frac{1}{p_2})(1 - \frac{1}{p_3}) \dots (1 - \frac{1}{p_k})$
- Sum of coprime to $n = n \times \frac{\phi(n)}{2}$
- If $n = 2^k$, $\phi(n) = 2^{k-1} = \frac{n}{2}$
- For a & b , $\phi(ab) = \phi(a)\phi(b) \cdot \frac{d}{\phi(d)}$
- $\phi(ip) = p\phi(i)$ whenever p is a prime and it divides i
- The number of a ($1 \leq a \leq N$) such that $\gcd(a, N) = d$ is $\phi(\frac{N}{d})$
- If $n > 2$, $\phi(n)$ is always even
- Sum of gcd, $\sum_{i=1}^n \gcd(i, n) = \sum_{d|n} d\phi(\frac{n}{d})$
- Sum of lcm, $\sum_{i=1}^n \operatorname{lcm}(i, n) = \frac{n}{2} (\sum_{d|n} (d\phi(d)) + 1)$
- $\phi(1) = 1$ and $\phi(2) = 1$ which two are only odd ϕ
- $\phi(3) = 2$ and $\phi(4) = 2$ and $\phi(6) = 2$ which three are only prime ϕ
- Find minimum n such that $\frac{\phi(n)}{n}$ is maximum - Multiple of small primes - $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \dots$

4.2.6 Mobius function

$\mu(1) = 1$. $\mu(n) = 0$, if n is not squarefree. $\mu(n) = (-1)^s$, if n is the product of s distinct primes. Let f, F be functions on positive integers. If for all $n \in \mathbb{N}$, $F(n) = \sum_{d|n} f(d)$, then $f(n) = \sum_{d|n} \mu(d)F(\frac{n}{d})$, and vice versa. $\phi(n) = \sum_{d|n} \mu(d)\frac{n}{d}$. $\sum_{d|n} \mu(d) = 1$. If f is multiplicative, then $\sum_{d|n} \mu(d)f(d) = \prod_{p|n} (1 - f(p))$, $\sum_{d|n} \mu(d)^2 f(d) = \prod_{p|n} (1 + f(p))$.

$$\sum_{i=1}^n \sum_{j=1}^n [\gcd(i, j) = 1] = \sum_{k=1}^n \mu(k) \left\lfloor \frac{n}{k} \right\rfloor^2$$

$$\sum_{i=1}^n \sum_{j=1}^n \gcd(i, j) = \sum_{k=1}^n k \sum_{l=1}^{\lfloor \frac{n}{k} \rfloor} \mu(l) \left\lfloor \frac{n}{kl} \right\rfloor^2$$

$$\sum_{i=1}^n \sum_{j=1}^n \gcd(i, j) = \sum_{k=1}^n \left(\frac{\lfloor \frac{n}{k} \rfloor (1 + \lfloor \frac{n}{k} \rfloor)}{2} \right)^2 \sum_{d|k} \mu(d) \frac{k}{d}$$

4.2.7 Legendre symbol

If p is an odd prime, $a \in \mathbb{Z}$, then $\left(\frac{a}{p}\right)$ equals 0, if $p|a$; 1 if a is a quadratic residue modulo p ; and -1 otherwise. Euler's criterion: $\left(\frac{a}{p}\right) = a^{\left(\frac{p-1}{2}\right)} \pmod{p}$.

4.2.8 Jacobi symbol

If $n = p_1^{a_1} \cdots p_k^{a_k}$ is odd, then $\left(\frac{a}{n}\right) = \prod_{i=1}^k \left(\frac{a}{p_i}\right)^{a_i}$.

4.2.9 Primitive roots

If the order of g modulo m ($\min n > 0: g^n \equiv 1 \pmod{m}$) is $\phi(m)$, then g is called a primitive root. If Z_m has a primitive root, then it has $\phi(\phi(m))$ distinct primitive roots. Z_m has a primitive root iff m is one of $2, 4, p^k, 2p^k$, where p is an odd prime. If Z_m has a primitive root g , then for all a coprime to m , there exists unique integer $i = \text{ind}_g(a)$ modulo $\phi(m)$, such that $g^i \equiv a \pmod{m}$. $\text{ind}_g(a)$ has logarithm-like properties: $\text{ind}(1) = 0$, $\text{ind}(ab) = \text{ind}(a) + \text{ind}(b)$.

If p is prime and a is not divisible by p , then congruence $x^n \equiv a \pmod{p}$ has $\gcd(n, p-1)$ solutions if $a^{(p-1)/\gcd(n, p-1)} \equiv 1 \pmod{p}$, and no solutions otherwise.

4.2.10 Discrete logarithm problem

Find x from $a^x \equiv b \pmod{m}$. Can be solved in $O(\sqrt{m})$ time and space with a meet-in-the-middle trick. Let $n = \lceil \sqrt{m} \rceil$, and $x = ny - z$. Equation becomes $a^{ny} \equiv ba^z \pmod{m}$. Precompute all values that the RHS can take for $z = 0, 1, \dots, n-1$, and brute force y on the LHS, each time checking whether there's a corresponding value for RHS.

4.2.11 Pythagorean triples

Integer solutions of $x^2 + y^2 = z^2$ All relatively prime triples are given by: $x = 2mn, y = m^2 - n^2, z = m^2 + n^2$ where $m > n, \gcd(m, n) = 1$ and $m \not\equiv n \pmod{2}$. All other triples are multiples of these. Equation $x^2 + y^2 = 2z^2$ is equivalent to $(\frac{x+y}{2})^2 + (\frac{x-y}{2})^2 = z^2$. The Pythagorean triples are uniquely generated by

$$a = k \cdot (m^2 - n^2), \quad b = k \cdot (2mn), \quad c = k \cdot (m^2 + n^2),$$

with $m > n > 0, k > 0, m \perp n$, and either m or n even.

4.2.12 Postage stamps/McNuggets problem

Let a, b be relatively-prime integers. There are exactly $\frac{1}{2}(a-1)(b-1)$ numbers *not* of form $ax + by$ ($x, y \geq 0$), and the largest is $(a-1)(b-1) - 1 = ab - a - b$.

4.2.13 Fermat's two-squares theorem

Odd prime p can be represented as a sum of two squares iff $p \equiv 1 \pmod{4}$. A product of two sums of two squares is a sum of two squares. Thus, n is a sum of two squares iff every prime of form $p = 4k+3$ occurs an even number of times in n 's factorization.

4.2.14 Modular Arithmetic

$$i \bmod j = i - j \left\lfloor \frac{i}{j} \right\rfloor$$

- HCN: 1e6(240), 1e9(1344), 1e12(6720), 1e14(17280), 1e15(26880), 1e16(41472)

$$\gcd(a, b, c, d, \dots) = \gcd(a, b - a, c - b, d - c, \dots)$$

$$\gcd(a + k, b + k, c + k, d + k, \dots) = \gcd(a + k, b - a, c - b, d - c, \dots)$$

- Primitive root exists iff $n = 1, 2, 4, p^k, 2 \times p^k$, where p is an odd prime.

- If primitive root exists, there are $\phi(\phi(n))$ primitive roots of n .

- The numbers from 1 to n have in total $O(n \log \log n)$ unique prime factors.

- $x \equiv r_1 \pmod{m_1}$ and $x \equiv r_2 \pmod{m_2}$ has a solution iff $\gcd(m_1, m_2) | (r_1 - r_2)$

$$ca \equiv cb \pmod{m} \iff a \equiv b \pmod{\frac{m}{\gcd(m, c)}}$$

$$ax \equiv b \pmod{m} \text{ has a solution } \iff \gcd(a, m) | b$$

- If $ax \equiv b \pmod{m}$ has a solution, then it has $\gcd(a, m)$ solutions and they are separated by $\frac{m}{\gcd(a, m)}$

$$ax \equiv 1 \pmod{m} \text{ has a solution or } a \text{ is invertible } \pmod{m} \iff \gcd(a, m) = 1$$

$$x^2 \equiv 1 \pmod{p} \text{ then } x \equiv \pm 1 \pmod{p}$$

$$\text{When } p \equiv 3 \pmod{4}, \text{ solution of } x^2 \equiv a \pmod{p} \text{ is } x \equiv \pm a^{\frac{p+1}{4}} \pmod{p}$$

$$\text{When } p \equiv 5 \pmod{8}, \text{ solution is } x \equiv a^{\frac{p+3}{8}} \pmod{p} \text{ or } x \equiv 2^{\frac{p-1}{4}} a^{\frac{p+3}{8}} \pmod{p}$$

4.2.15 Wilson's Theorem

Wilson's Theorem states that an integer $p > 1$ is a prime number if and only if:

$$(p-1)! \equiv -1 \pmod{p}$$

Equivalently, in positive modulo representation:

$$(p-1)! \equiv p-1 \pmod{p}$$

Competitive Programming Applications: While not practical for general primality testing due to the $O(p)$ factorial computation, Wilson's Theorem is extremely useful for calculating factorials close to p modulo p .

If you need to find $(p-k)! \pmod{p}$ for a small k , you can use Wilson's Theorem to compute it in $O(k \log p)$ time (using modular inverse) instead of $O(p)$ time.

Useful Corollaries: By expanding $(p-1)! = (p-1)(p-2)!$ and dividing out terms, we get:

$$(p-1)! \equiv -1 \pmod{p}$$

$$(p-2)! \equiv 1 \pmod{p}$$

$$(p-k)! \equiv \frac{(-1)^k}{(k-1)!} \pmod{p} \quad (\text{for } 1 \leq k \leq p)$$

4.3 Combinatorics**4.3.1 Permutations and Combinations Formulas**

$$\text{Permutation: } {}^n P_r = \frac{n!}{(n-r)!}$$

$$\text{Combination: } {}^n C_r = \frac{n!}{r!(n-r)!}$$

4.3.2 Factorial

$\frac{n}{n!}$	1	2	3	4	5	6	7	8	9	10
$\frac{n}{n!}$	1	2	6	24	120	720	5040	40320	362880	3628800
$\frac{n}{n!}$	11	11	12	12	13	14	15	16	17	17
$\frac{n}{n!}$	4.0e7	4.8e8	6.2e9	8.7e10	1.3e12	2.1e13	3.6e14	150	171	
$\frac{n}{n!}$	2e18	2e25	3e32	8e47	3e64	9e157	6e262	>DBL_MAX		

4.3.3 Binomial Coefficient

$$\text{Factoring in: } \binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}$$

$$\text{Sum over } k: \sum_{k=0}^n \binom{n}{k} = 2^n$$

$$\text{Alternating sum: } \sum_{k=0}^n (-1)^k \binom{n}{k} = 0$$

$$\text{Even and odd sum: } \sum_{k=0}^n \binom{n}{2k} = \sum_{k=0}^n \binom{n}{2k+1} = 2^{n-1}$$

- The Hockey Stick Identity:

$$\text{– (Left to right) Sum over } n \text{ and } k: \sum_{k=0}^m \binom{n+k}{k} = \binom{n+m+1}{m}$$

$$\text{– (Right to left) Sum over } n: \sum_{m=k}^n \binom{m}{k} = \binom{n+1}{k+1}$$

$$\text{Sum of the squares: } \sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$$

$$\text{Weighted sum: } \sum_{k=1}^n k \binom{n}{k} = n 2^{n-1}$$

$$\text{Connection with the Fibonacci numbers: } \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} = F_{n+1}$$

$$\text{Vandermonde's Identity: } \sum_{i=0}^k \binom{m}{i} \binom{n}{k-i} = \binom{m+n}{k}$$

- If $f(n, k) = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{k}$, Then $f(n+1, k) = 2f(n, k) - \binom{n}{k}$ [For multiple $f(n, k)$ queries, use Mo's algo]

Lucas Theorem

$$\binom{m}{n} \equiv \prod_{i=0}^k \binom{m_i}{n_i} \pmod{p}$$

- $\binom{m}{n}$ is divisible by p if and only if at least one of the base- p digits of n is greater than the corresponding base- p digit of m .

- The number of entries in the n -th row of Pascal's triangle that are not divisible by $p = \prod_{i=0}^k (n_i + 1)$

- All entries in the $(p^k - 1)$ -th row are not divisible by p .

$$\binom{n}{m} \equiv \left\lfloor \frac{n}{p} \right\rfloor \pmod{p}$$

4.3.4 Cycles

Let $g_S(n)$ be the number of n -permutations whose cycle lengths all belong to the set S . Then

$$\sum_{n=0}^{\infty} g_S(n) \frac{x^n}{n!} = \exp \left(\sum_{n \in S} \frac{x^n}{n} \right)$$

4.3.5 Derangements

Permutations of a set such that none of the elements appear in their original position.

$$D(n)=(n-1)(D(n-1)+D(n-2))=nD(n-1)+(-1)^n=\left\lfloor \frac{n!}{e} \right\rfloor$$

4.3.6 Burnside's lemma

Given a group G of symmetries and a set X , the number of elements of X up to symmetry equals

$$\frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where X^g are the elements fixed by g ($g.x = x$).
If $f(n)$ counts “configurations” (of some sort) of length n , we can ignore rotational symmetry using $G = \mathbb{Z}_n$ to get

$$g(n)=\frac{1}{n}\sum_{k=0}^{n-1}f(\gcd(n,k))=\frac{1}{n}\sum_{k|n}f(k)\phi(n/k)$$

4.3.7 Multinomial Coefficients

$$\frac{(C_1+C_2+\cdots+C_N)!}{C_1!C_2!\cdots C_N!}=\binom{C_1}{C_1}\binom{C_1+C_2}{C_2}\binom{C_1+C_2+C_3}{C_3}\cdots\binom{C_1+\cdots+C_N}{C_N}$$

4.3.8 Partitions and Subsets

Partition function Number of ways of writing n as a sum of positive integers, disregarding the order of the summands.

$$p(0)=1, \, p(n)=\sum_{k \in \mathbb{Z} \setminus \{0\}} (-1)^{k+1} p(n-k(3k-1)/2)$$

$$p(n) \sim \frac{0.145}{n} \exp(2.56 \sqrt{n})$$

n	0	1	2	3	4	5	6	7	8	9	20	50	100
$p(n)$	1	1	2	3	5	7	11	15	22	30	627	~2e5	~2e8

Partition Number Calculation

- Time Complexity: $O(n\sqrt{n})$

```
for (int i = 1; i <= n; ++i) {
    pent[2 * i - 1] = i * (3 * i - 1) / 2;
    pent[2 * i] = i * (3 * i + 1) / 2;
}
p[0] = 1;
for (int i = 1; i <= n; ++i) {
    p[i] = 0;
    for (int j = 1, k = 0; pent[j] <= i; ++j) {
        if (k < 2) p[i] = add(p[i], p[i - pent[j]]);
        else p[i] = sub(p[i], p[i - pent[j]]); ++k, k &= 3;
    }
}
```

The number of partitions of a positive integer n into exactly k parts equals the number of partitions of n whose largest part equals k

$$p_k(n)=p_k(n-k)+p_{k-1}(n-1)$$

2nd Kaplansky's Lemma

The number of ways of selecting k objects, no two consecutive, from n labelled objects arrayed in a circle is $\frac{n}{k}\binom{n-k-1}{k-1}=\frac{n}{n-k}\binom{n-k}{k}$

Distinct Objects into Distinct Bins

- n distinct objects into r distinct bins $=r^n$
- Among n distinct objects, exactly k of them into r distinct bins $=\binom{n}{k}r^k$
- n distinct objects into r distinct bins such that each bin contains at least one object $=\sum_{i=0}^r (-1)^i \binom{r}{i} (r-i)^n$

4.3.9 General purpose numbers

Eulerian numbers Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element.

$$E(n,k)=(n-k)E(n-1,k-1)+(k+1)E(n-1,k)$$

$$E(n,0)=E(n,n-1)=1$$

$$E(n,k)=\sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n$$

Bell numbers Total number of partitions of n distinct elements. $B(n)=1,1,2,5,15,52,203,877,4140,21147,\dots$ For p prime,

$$B(p^m+n)\equiv mB(n)+B(n+1)\pmod p$$

Bernoulli numbers $\sum_{j=0}^m \binom{m+1}{j} B_j = 0.$ $B_0=1, B_1=-\frac{1}{2}.$ $B_n=0$, for all odd $n \neq 1$.

Catalan numbers

$$C_n=\frac{1}{n+1}\binom{2n}{n}=\binom{2n}{n}-\binom{2n}{n+1}=\frac{(2n)!}{(n+1)!n!}$$

$$C_0=1, \, C_{n+1}=\frac{2(2n+1)}{n+2}C_n, \, C_{n+1}=\sum C_iC_{n-i}$$

- $C_n=1,1,2,5,14,42,132,429,1430,4862,16796,58786,\dots$
- sub-diagonal monotone paths in an $n \times n$ grid.
- strings with n pairs of parenthesis, correctly nested.
- binary trees with $n+1$ leaves (0 or 2 children).
- ordered trees with $n+1$ vertices.
- ways a convex polygon with $n+2$ sides can be cut into triangles by connecting vertices with straight lines.
- permutations of $[n]$ with no 3-term increasing subseq.
- Find the count of balanced parentheses sequences consisting of $n+k$ pairs of parentheses where the first k symbols are open brackets.

$$C_n^{(k)}=\frac{k+1}{n+k+1}\binom{2n+k}{n}$$

- Recursive formula of Catalan Numbers:

$$C_n^{(k)}=\frac{(2n+k-1)\cdot(2n+k)}{n\cdot(n+k+1)}C_{n-1}^{(k)}$$

Lucas Number Number of edge cover of a cycle graph C_n is L_n

$$L(n)=L(n-1)+L(n-2);L(0)=2,L(1)=1$$

4.3.10 Stars and Bars Theorems

- No Empty Bins Allowed: $\binom{N-1}{K-1}$
- Empty Bins Allowed: $\binom{N+K-1}{K-1}$

4.3.11 Ballot Theorem

Suppose that in an election, candidate A receives a votes and candidate B receives b votes, where $a \geq kb$ for some positive integer k . Compute the number of ways the ballots can be ordered so that A maintains more than k times as many votes as B throughout the counting of the ballots. The solution to the ballot problem is $\frac{a-kb}{a+b} \times \binom{a+b}{a}$

4.4 Sequences and Series

4.4.1 Fibonacci Number

- $k=A-B, F_A F_B = F_{k+1} F_A^2 + F_k F_A F_{A-1}$
- $\sum_{i=0}^n F_i^2 = F_{n+1} F_n$
- $\sum_{i=0}^n F_i F_{i+1} = F_{n+1}^2 - (-1)^n$
- $\sum_{i=0}^n F_i F_{i-1} = \sum_{i=0}^{n-1} F_i F_{i+1}$
- $\gcd(F_m, F_n) = F_{\gcd(m,n)}$
- $\gcd(F_n, F_{n+1}) = \gcd(F_n, F_{n+2}) = \gcd(F_{n+1}, F_{n+2}) = 1$

4.4.2 Sums

$$1+2+3+\cdots+n=\frac{n(n+1)}{2}$$
$$1+3+5+\cdots+(2n-1)=n^2$$
$$2+4+6+\cdots+2n=n(n+1)$$
$$1^2+2^2+3^2+\cdots+n^2=\frac{n(n+1)(2n+1)}{6}$$
$$1^3+2^3+3^3+\cdots+n^3=\frac{n^2(n+1)^2}{4}$$
$$1^4+2^4+3^4+\cdots+n^4=\frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

$$\sum_{i=1}^n i^m=\frac{1}{m+1}\left[(n+1)^{m+1}-1-\sum_{i=1}^n((i+1)^{m+1}-i^{m+1}-(m+1)i^m)\right]$$
$$\sum_{i=1}^{n-1} i^m=\frac{1}{m+1}\sum_{k=0}^m\binom{m+1}{k}B_k n^{m+1-k}$$
$$\sum_{k=0}^n kx^k=\frac{x-(n+1)x^{n+1}+nx^{n+2}}{(x-1)^2}$$

4.4.3 Progressions

Arithmetic Progression

- Sequence:** $a, a+d, a+2d, \dots, a+(n-1)d$
- Sum of first n terms:** $S_n=\frac{n}{2}[2a+(n-1)d]$

Geometric Progression

- Sequence:** $a, ar, ar^2, \dots, ar^{n-1}$
- Sum (for $r > 1$):** $S_n=\frac{a(r^n-1)}{r-1}$
- Sum (for $r < 1$):** $S_n=\frac{a(1-r^n)}{1-r}$

4.4.4 Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, (-\infty < x < \infty)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, (-1 < x \leq 1)$$

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{2x^3}{32} - \frac{5x^4}{128} + \dots, (-1 \leq x \leq 1)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots, (-\infty < x < \infty)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots, (-\infty < x < \infty)$$

$$(x+a)^{-n} = \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} x^k a^{-n-k}$$

Generating Function

$$1/(1-x) = 1 + x + x^2 + x^3 + \dots$$

$$1/(1-ax) = 1 + ax + (ax)^2 + (ax)^3 + \dots$$

$$1/(1-x)^2 = 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$1/(1-x)^3 = \binom{2}{2} + \binom{3}{2}x + \binom{4}{2}x^2 + \binom{5}{2}x^3 + \dots$$

$$1/(1-ax)^{k+1} = 1 + \binom{1+k}{k}(ax) + \binom{2+k}{k}(ax)^2 + \binom{3+k}{k}(ax)^3 + \dots$$

$$x(x+1)(1-x)^{-3} = 1 + x + 4x^2 + 9x^3 + 16x^4 + 25x^5 + \dots$$

$$e^x = 1 + x + x^2/2! + x^3/3! + x^4/4! + \dots$$

4.4.5 Logarithms**Change of Base Formula**

$$\log_a x = \frac{\log_b x}{\log_b a}$$

4.4.6 Inequalities

Titu's Lemma For positive reals $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$,

$$\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2} + \dots + \frac{a_n^2}{b_n} \geq \frac{(a_1 + a_2 + \dots + a_n)^2}{b_1 + b_2 + \dots + b_n}$$

Equality holds if and only if $a_i = kb_i$ for a non-zero real constant k .

4.5 Graph Theory**4.5.1 Coloring**

- The number of labeled undirected graphs with n vertices, $G_n = 2^{\binom{n}{2}}$
- The number of labeled directed graphs with n vertices, $G_n = 2^{n(n-1)}$
- The number of connected labeled undirected graphs with n vertices, $C_n = 2^{\binom{n}{2}} - \frac{1}{n} \sum_{k=1}^{n-1} k \binom{n}{k} 2^{\binom{n-k}{2}} C_k = 2^{\binom{n}{2}} - \sum_{k=1}^{n-1} \sum_{k=1}^{n-1} \binom{n-1}{k-1} 2^{\binom{n-k}{2}} C_k$
- The number of k -connected labeled undirected graphs with n vertices, $D[n][k] = \sum_{s=1}^n \binom{n-1}{s-1} C_s D[n-s][k-1]$
- Cayley's formula: the number of trees on n labeled vertices = the number of spanning trees of a complete graph with n labeled vertices = n^{n-2}
- Number of ways to color a graph using k colors such that no two adjacent nodes have the same color:
 - Complete graph = $k(k-1)(k-2)\dots(k-n+1)$
 - Tree = $k(k-1)^{n-1}$
 - Cycle = $(k-1)^n + (-1)^n(k-1)$
- Number of trees with n labeled nodes: n^{n-2}

4.5.2 Classical Problem

$F(n, k)$ = number of ways to color n objects using exactly k colors

Let $G(n, k)$ be the number of ways to color n objects using no more than k colors.

Then, $F(n, k) = G(n, k) - \binom{k}{1}G(n, k-1) + \binom{k}{2}G(n, k-2) - \binom{k}{3}G(n, k-3) \dots$

Determining $G(n, k)$: Suppose, we are given a $1 \times n$ grid. Any two adjacent cells can not have the same color. Then, $G(n, k) = k(k-1)^{n-1}$

If no such condition on adjacent cells: Then, $G(n, k) = k^n$

4.5.3 Matching Formula**Normal Graph**

MM + MEC = n (excluding vertex), IS + VC = G , MIS + MVC = G

Bipartite Graph

MIS = n - MBM, MVC = MBM, MEC = n - MBM

4.6 Strings and Bitwise**4.6.1 String**

- If the sum of length of some strings is N , there can be at most $\sqrt{N}$ distinct lengths.
- A Text can have at most $O(N\sqrt{N})$ distinct substrings that match with given patterns where the sum of the length of the given patterns is N .
- Period = $n \pmod{n - \text{pi.back}()} = 0$? $n - \text{pi.back}() : n$
- The first (period) cyclic rotations of a string are distinct. Further cyclic rotations repeat the previous strings.
- S is a palindrome if and only if its period is a palindrome.
- If S and T are palindromes, then the periods of S & T are the same if and only if $S + T$ is a palindrome.

4.6.2 Tree Hashing

$f(u) = sz[u] \times \sum_{i=0} f(v) \times p^i$; $f(v)$ are sorted

$f(child) = 1$

4.6.3 Bit

- $(a \oplus b)$ and $(a + b)$ have the same parity
- $(a + b) = (a \oplus b) + 2(a \& b)$
- $\gcd(a, b) \leq |a - b| \leq (a \oplus b)$

4.6.4 Convolution

- Hamming Distance: Replace 0 with -1

- SQRT Decomposition: Find block size, $B = \sqrt{8n}$

4.6.5 Permutation Adjacency Difference

To maximize the sum of adjacent differences of a permutation, it is necessary and sufficient to place the smallest half numbers in odd positions and the greatest half numbers in even positions. Or, vice versa.

4.7 Game Theory**4.7.1 Grundy numbers**

For a two-player, normal-play (last to move wins) game on a graph (V, E) : $G(x) = \text{mex}(\{G(y) : (x, y) \in E\})$, where $\text{mex}(S) = \min\{n \geq 0 : n \notin S\}$. x is losing iff $G(x) = 0$.

4.7.2 Sums of games

- Player chooses a game and makes a move in it*: Grundy number of a position is xor of grundy numbers of positions in summed games.
- Player chooses a non-empty subset of games (possibly, all) and makes moves in all of them*: A position is losing iff each game is in a losing position.
- Player chooses a proper subset of games (not empty and not all), and makes moves in all chosen ones*: A position is losing iff grundy numbers of all games are equal.
- Player must move in all games, and loses if can't move in some game*: A position is losing if any of the games is in a losing position.

4.7.3 Misère Nim

A position with pile sizes $a_1, a_2, \dots, a_n \geq 1$, not all equal to 1, is losing iff $a_1 \oplus a_2 \oplus \dots \oplus a_n = 0$ (like in normal nim.) A position with n piles of size 1 is losing iff n is *odd*.